

Series and Progressions

CSC 1105 Mathematics for Computer Science

Denish Azamuke, PhD

Office: CoCIS, Block B, Level 3

August 24, 2026

Lecture Outline

- 1 Sequences, series, and summation notation
- 2 Arithmetic and geometric progressions
- 3 Special progressions and combined series
- 4 Taylor and Maclaurin series
- 5 Sequence approximation using interpolation

Learning Outcomes

By the end of this lecture, you should be able to:

- distinguish between a sequence, a progression, and a series;
- derive and apply formulae for arithmetic and geometric sums;
- simplify sums using special progressions;
- approximate functions using Taylor and Maclaurin polynomials;
- construct interpolation polynomials from discrete data points.

Part 1/2

Sequences and Series

Sequence

A sequence is an ordered list of numbers:

$$a_1, a_2, a_3, \dots, a_n, \dots$$

where a_n is the n th term.

Series

A series is the sum of terms of a sequence:

$$S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k.$$

Why Series Matter in Computer Science

Series and progressions appear in many computing problems:

- estimating running time, for example $1 + 2 + \cdots + n$;
- analysing loops and recursive algorithms;
- modelling growth, savings, and compound interest;
- approximating functions for numerical computation;
- reconstructing unknown values from sampled data.

Arithmetic Progression (AP)

An arithmetic progression has a constant difference d between consecutive terms:

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \dots$$

Nth Term

$$a_n = a + (n - 1)d.$$

Sum of First n Terms

$$S_n = \frac{n}{2} (2a + (n - 1)d) = \frac{n}{2} (a + l),$$

where l is the last term.

AP Example

Find the sum of the first 50 positive integers.

$$1 + 2 + 3 + \cdots + 50$$

Here $a = 1$, $d = 1$, and $n = 50$.

$$S_{50} = \frac{50}{2} (2(1) + (50 - 1)(1)) = 25(51) = 1275.$$

General Result

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Geometric Progression (GP)

A geometric progression has a constant ratio r between consecutive terms:

$$a, \quad ar, \quad ar^2, \quad ar^3, \dots$$

Nth Term

$$a_n = ar^{n-1}.$$

Sum of First n Terms

$$S_n = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1.$$

Infinite Geometric Series

For $|r| < 1$, the infinite geometric series converges:

$$S_{\infty} = a + ar + ar^2 + ar^3 + \cdots = \frac{a}{1 - r}.$$

Example

$$3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \cdots$$

Here $a = 3$ and $r = \frac{1}{2}$.

$$S_{\infty} = \frac{3}{1 - \frac{1}{2}} = 6.$$

Application: Compound Interest

Suppose UGX P is invested at interest rate r per period. After n periods, the value is

$$A = P(1 + r)^n.$$

Example

If UGX 1,000,000 is invested at 10% per year for 3 years,

$$A = 1,000,000(1.10)^3 = 1,331,000.$$

This is a geometric progression because each period multiplies the previous amount by $1 + r$.

Application: Regular Savings

If UGX a is saved every period and each saving grows by a factor r , the accumulated amount is often a GP sum:

$$a + ar + ar^2 + \cdots + ar^{n-1}.$$

$$S_n = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1.$$

Interpretation

This type of model is useful for savings plans, loan repayments, and repeated investments.

Proof by Induction

Many computer science problems involve structures or processes that grow **one step at a time**. Mathematical induction provides a natural way to prove that such processes work correctly for **any size of input**.

- **Algorithm correctness:** prove that an algorithm works for inputs of arbitrary size.
- **Recursive algorithms:** verify programs that solve a problem using smaller instances of the same problem.
- **Loops and iterations:** prove properties that remain true after every iteration.
- **Data structures:** establish properties of recursively defined structures such as trees and linked lists.
- **Complexity analysis:** prove formulas for running time, memory usage, and recurrence relations.

Proof by Induction: Basic Structure

To prove that a statement $P(n)$ is true for all $n \in \mathbb{N}$:

- ① **Base case:** show that $P(1)$ is true.
- ② **Induction assumption:** assume $P(k)$ is true for some $k \in \mathbb{N}$.
- ③ **Induction step:** prove that $P(k + 1)$ is true.

Conclusion

If all three steps hold, then $P(n)$ is true for every positive integer n .

Induction Example

Prove that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Base case: for $n = 1$,

$$1 = \frac{1(1+1)}{2} = 1.$$

Assume true for $n = k$:

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Induction Example Continued

For $n = k + 1$,

$$\begin{aligned}1 + 2 + \cdots + k + (k + 1) &= \frac{k(k + 1)}{2} + (k + 1) \\&= (k + 1) \left(\frac{k}{2} + 1 \right) \\&= \frac{(k + 1)(k + 2)}{2}.\end{aligned}$$

This is the required formula with $n = k + 1$. Therefore,

$$1 + 2 + 3 + \cdots + n = \frac{n(n + 1)}{2}.$$

Special Progressions

The following sums are frequently used:

$$\begin{aligned}1 + 2 + 3 + \cdots + n &= \frac{n(n+1)}{2}, \\1^2 + 2^2 + 3^2 + \cdots + n^2 &= \frac{n(n+1)(2n+1)}{6}, \\1^3 + 2^3 + 3^3 + \cdots + n^3 &= \left(\frac{n(n+1)}{2}\right)^2.\end{aligned}$$

Key idea

Many difficult-looking series can be rewritten as combinations of these special progressions.

Deriving Sum of Squares

We want to derive a formula for

$$1^2 + 2^2 + 3^2 + \cdots + n^2$$

or, using summation notation,

$$\sum_{k=1}^n k^2.$$

Target Formula

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Step 1: Start with a Useful Identity

Consider the expression

$$n^3 - (n - 1)^3.$$

First expand:

$$(n - 1)^3 = n^3 - 3n^2 + 3n - 1.$$

Therefore,

$$\begin{aligned} n^3 - (n - 1)^3 &= n^3 - (n^3 - 3n^2 + 3n - 1) \\ &= 3n^2 - 3n + 1. \end{aligned}$$

So,

$$n^3 - (n - 1)^3 = 3n^2 - 3n + 1$$

Step 2: Replace n by k

Since the identity is true for any positive integer, we write

$$k^3 - (k - 1)^3 = 3k^2 - 3k + 1.$$

Now sum both sides from $k = 1$ to $k = n$:

$$\sum_{k=1}^n (k^3 - (k - 1)^3) = \sum_{k=1}^n (3k^2 - 3k + 1).$$

Why do this?

The left side will simplify by cancellation, while the right side contains $\sum k^2$, which is what we want.

Step 3: The Left Side Telescopes

Expand the left side:

$$\begin{aligned}\sum_{k=1}^n (k^3 - (k-1)^3) &= (1^3 - 0^3) + (2^3 - 1^3) \\ &\quad + (3^3 - 2^3) + \cdots + (n^3 - (n-1)^3).\end{aligned}$$

Most terms cancel:

$$1^3 - 0^3 + 2^3 - 1^3 + 3^3 - 2^3 + \cdots + n^3 - (n-1)^3.$$

Only the first negative term and the last positive term remain:

$$\boxed{\sum_{k=1}^n (k^3 - (k-1)^3) = n^3}$$

Step 4: Split the Right Side

From the identity,

$$n^3 = \sum_{k=1}^n (3k^2 - 3k + 1).$$

Using summation rules:

$$n^3 = 3 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + \sum_{k=1}^n 1.$$

But

$$\sum_{k=1}^n 1 = n.$$

So,

Step 5: Use the Formula for $\sum k$

Recall that

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Substitute into

$$n^3 = 3 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + n.$$

This gives

$$n^3 = 3 \sum_{k=1}^n k^2 - 3 \left(\frac{n(n+1)}{2} \right) + n.$$

Now isolate the sum of squares:

Step 6: Simplify

Start with

$$3 \sum_{k=1}^n k^2 = n^3 + \frac{3n(n+1)}{2} - n.$$

Put everything over 2:

$$3 \sum_{k=1}^n k^2 = \frac{2n^3 + 3n(n+1) - 2n}{2}.$$

Expand:

$$3 \sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + 3n - 2n}{2}.$$

$$3 \sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + n}{2}.$$

Step 7: Factor and Divide

Factor the numerator:

$$2n^3 + 3n^2 + n = n(2n^2 + 3n + 1).$$

Then factor the quadratic:

$$2n^2 + 3n + 1 = (n + 1)(2n + 1).$$

So,

$$3 \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{2}.$$

Divide both sides by 3:

$$\boxed{\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}}$$

Check

Let $n = 4$.

Using direct addition:

$$1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30.$$

Using the formula:

$$\begin{aligned}\sum_{k=1}^4 k^2 &= \frac{4(4+1)(2(4)+1)}{6} \\ &= \frac{4 \cdot 5 \cdot 9}{6} = 30.\end{aligned}$$

Conclusion

Both methods give the same answer, so the formula works for $n = 4$.

Combined Progressions

Some progressions are products or combinations of simpler ones.

Example

Find the sum of the first n terms of

$$1 \cdot 3 + 3 \cdot 5 + 5 \cdot 7 + \dots$$

The n th term is

$$a_n = (2n - 1)(2n + 1) = 4n^2 - 1.$$

Therefore,

$$S_n = \sum_{k=1}^n (4k^2 - 1) = 4 \sum_{k=1}^n k^2 - \sum_{k=1}^n 1.$$

Note: Finding the n th Term

The terms are

$$1 \cdot 3, \quad 3 \cdot 5, \quad 5 \cdot 7, \quad 7 \cdot 9, \quad \dots$$

Observe the pattern:

n	1	2	3
first factor	1	3	5
second factor	3	5	7

The first factors form the odd numbers:

$$1, 3, 5, \dots = 2n - 1.$$

The second factor is always the next odd number:

$$3, 5, 7, \dots = 2n + 1.$$

Therefore, the n th term is

$$a_n = (2n - 1)(2n + 1) = 4n^2 - 1.$$

Combined Progressions Continued

Using the special sums,

$$\begin{aligned} S_n &= 4 \cdot \frac{n(n+1)(2n+1)}{6} - n \\ &= \frac{2n(n+1)(2n+1)}{3} - n \\ &= \frac{n(4n^2 + 6n - 1)}{3}. \end{aligned}$$

Answer

$$S_n = \frac{n(4n^2 + 6n - 1)}{3}.$$

Polynomial Term Example

Given

$$a_n = n(n+1)(n+2),$$

find the sum of the first n terms.

First expand:

$$a_n = n^3 + 3n^2 + 2n.$$

Then

$$S_n = \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + 2 \sum_{k=1}^n k.$$

Polynomial Term Example Continued

Substitute the standard sums:

$$S_n = \left(\frac{n(n+1)}{2} \right)^2 + 3 \cdot \frac{n(n+1)(2n+1)}{6} + 2 \cdot \frac{n(n+1)}{2}.$$

Simplified Form

$$S_n = \frac{n(n+1)(n+2)(n+3)}{4}.$$

Exercise: Combined Progressions

For each n th term, find the expression for the sum of the first n terms.

① $a_n = n + 2$

② $a_n = n^2 - 2n$

③ $a_n = (2n + 1)(n - 1)$

④ $a_n = (n + 2)(1 - n)$

⑤ $a_n = (n + 1)(n + 2)(n + 3)$

Hint: expand first, then use the special sums.

Questions?

Part 2/2

Taylor S



Taylor Series

